Finding character frequency in given string:

bahu bali

a 2 times

b 2 times

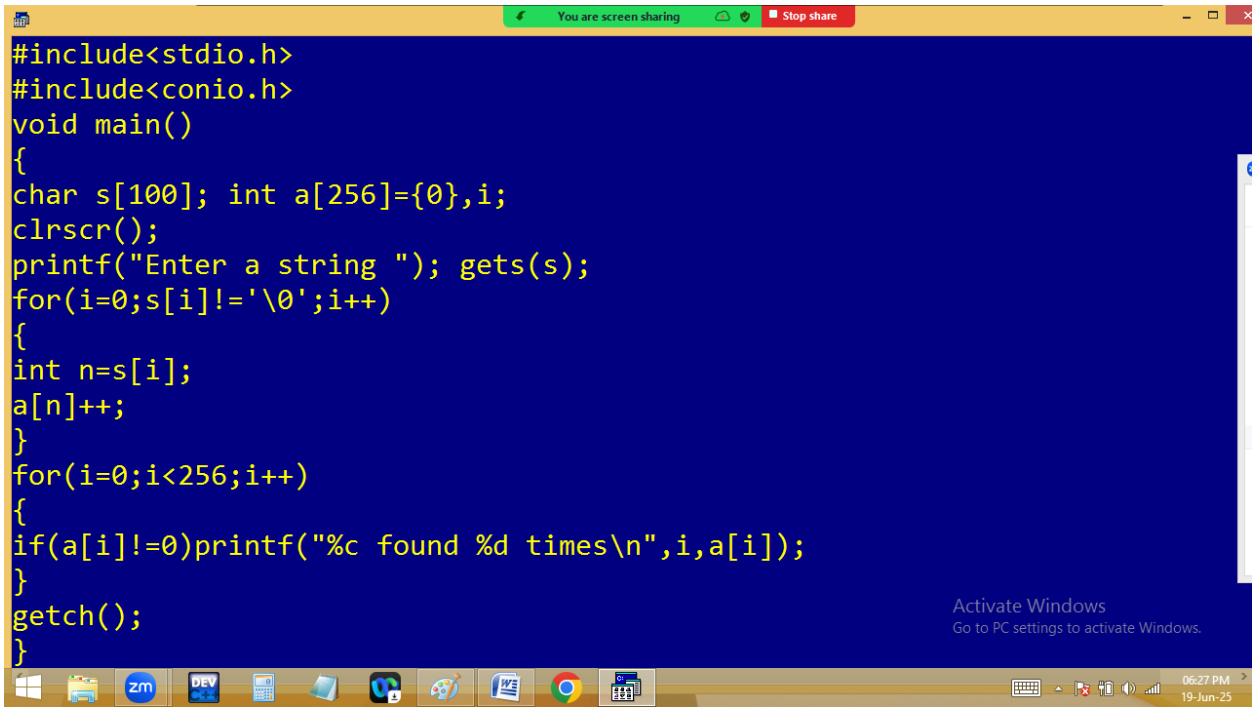
h 1 time

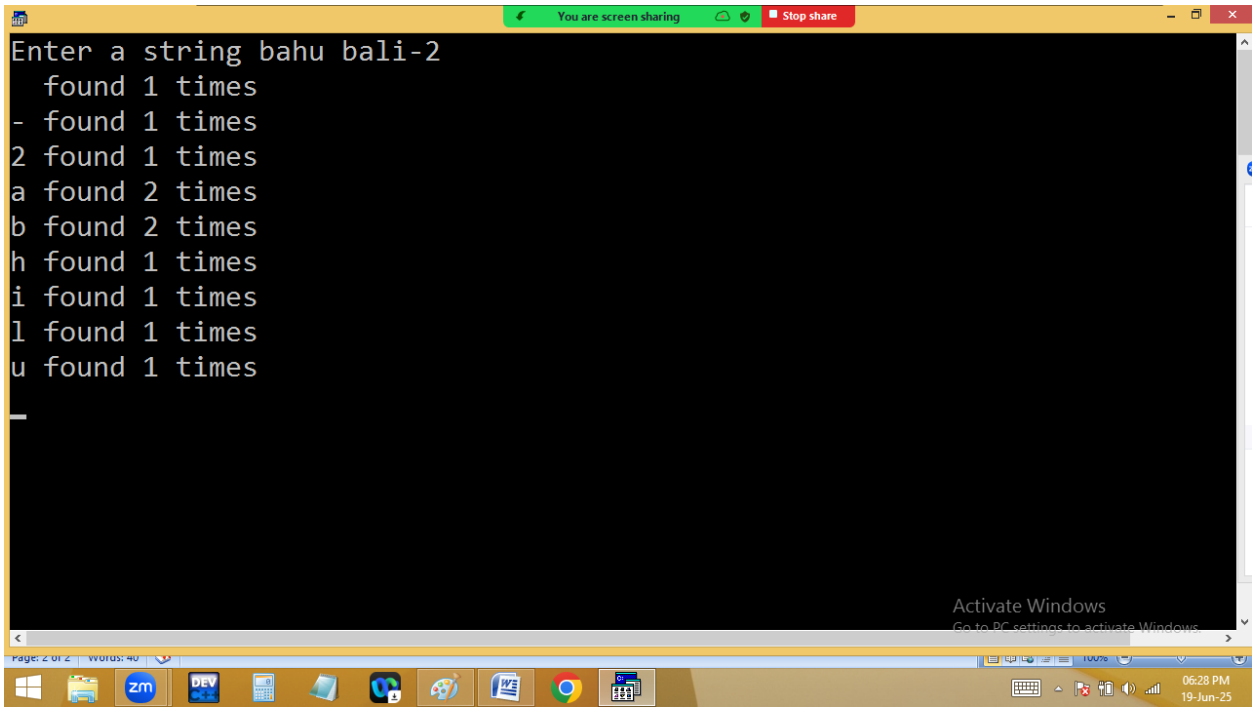
i 1 time

l 1 time

Space 1 time

U 1 time

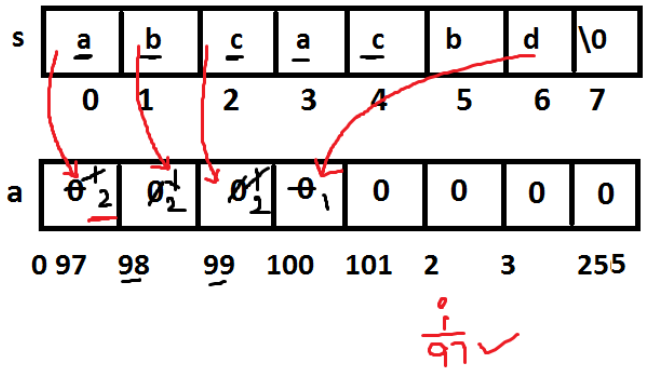

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Stop share
#include<stdio.h>
#include<conio.h>
void main()
{
char s[100]; int a[256]={0},i;
clrscr();
printf("Enter a string "); gets(s);
for(i=0;s[i]!='\0';i++)
{
int n=s[i];
a[n]++;
}
for(i=0;i<256;i++)
{
if(a[i]!=0)printf("%c found %d times\n",i,a[i]);
}
getch();
}
Activate Windows
Go to PC settings to activate Windows.
06:27 PM
19-Jun-25




```
for( i=0; s[i]!='\0';i++)
{
int n=s[i];
a[n]++;
}

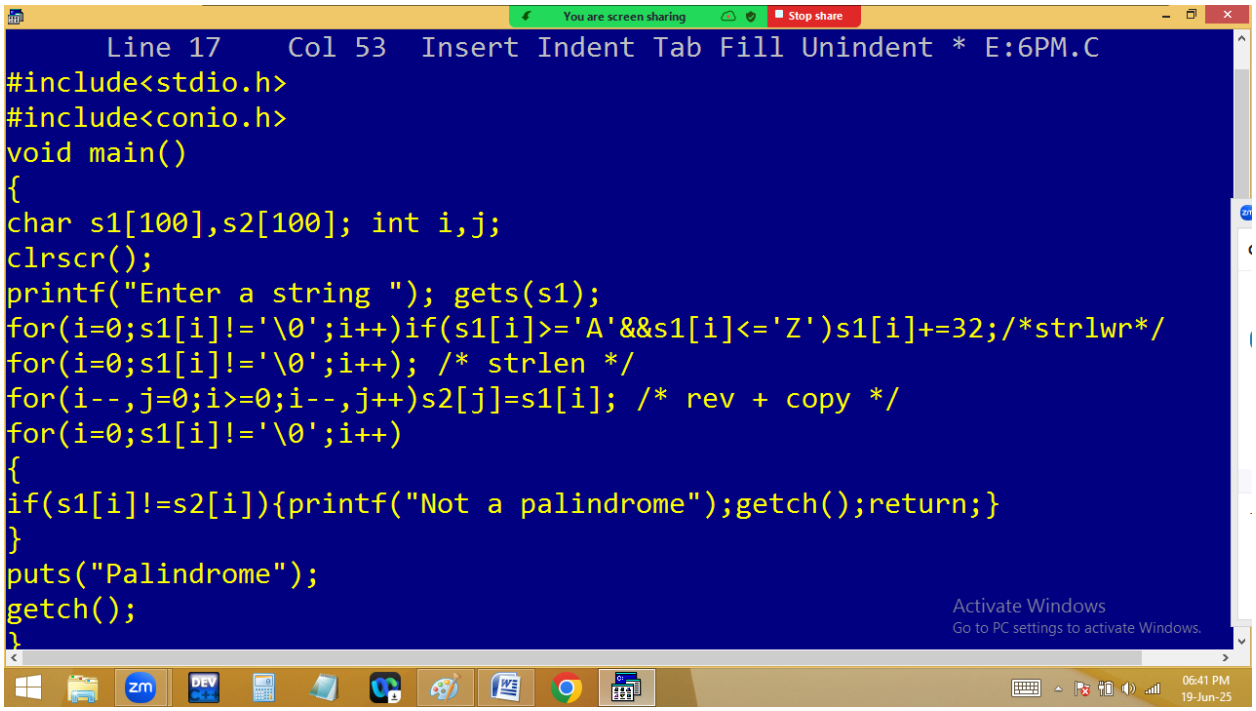
for(i=0;i<256;i++)
if(a[i]!=0)p("%c found %d times\n",i,a[i]);
```

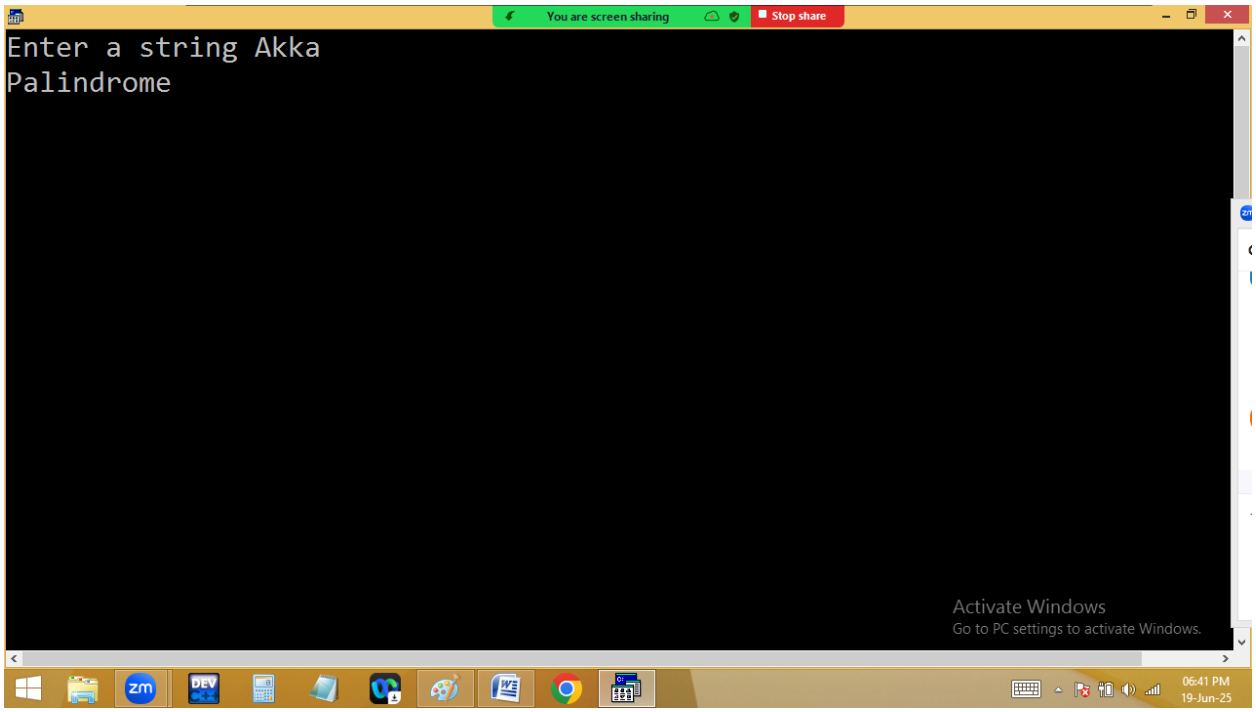
a found 2 times
b found 2 times
c found 2 times
d found 1 time

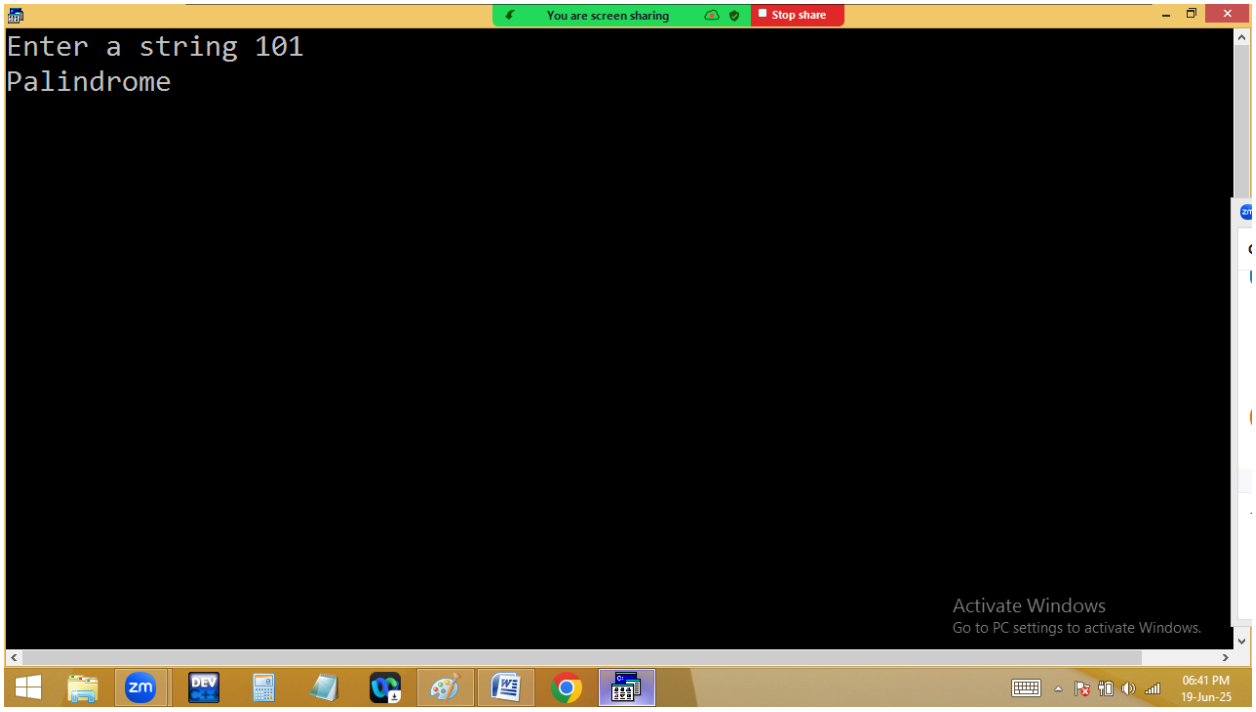


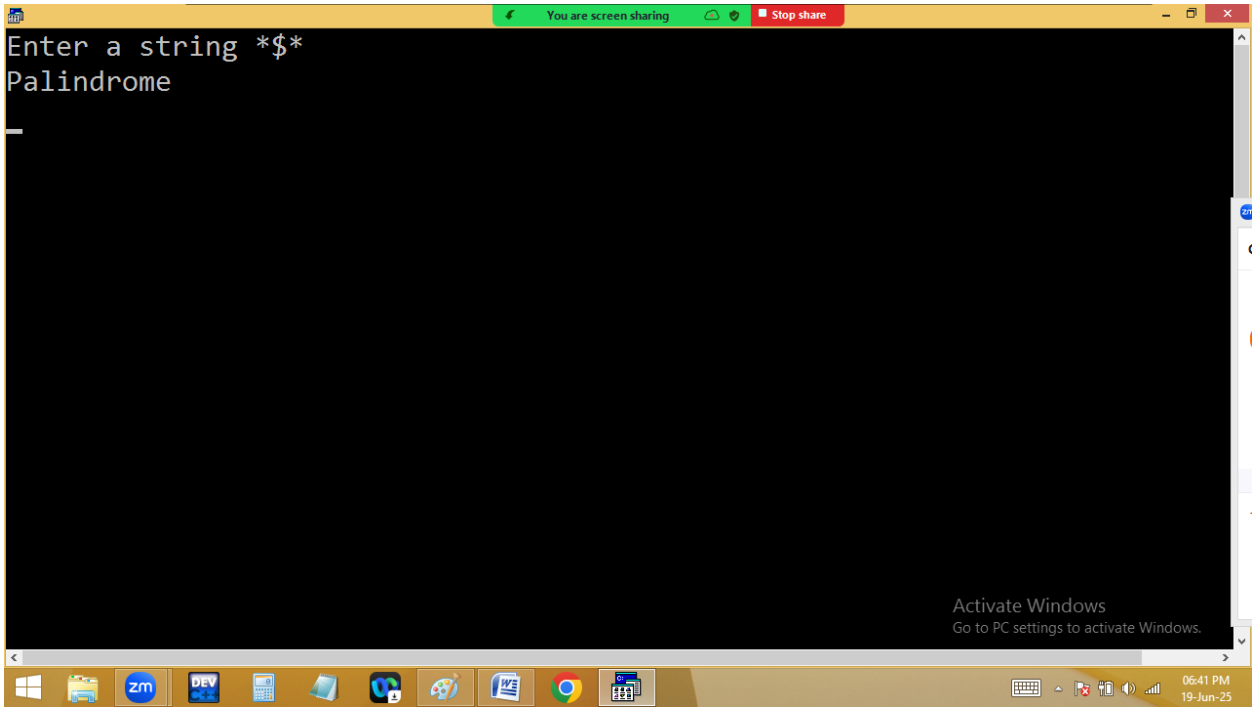
2. Finding palindrome string

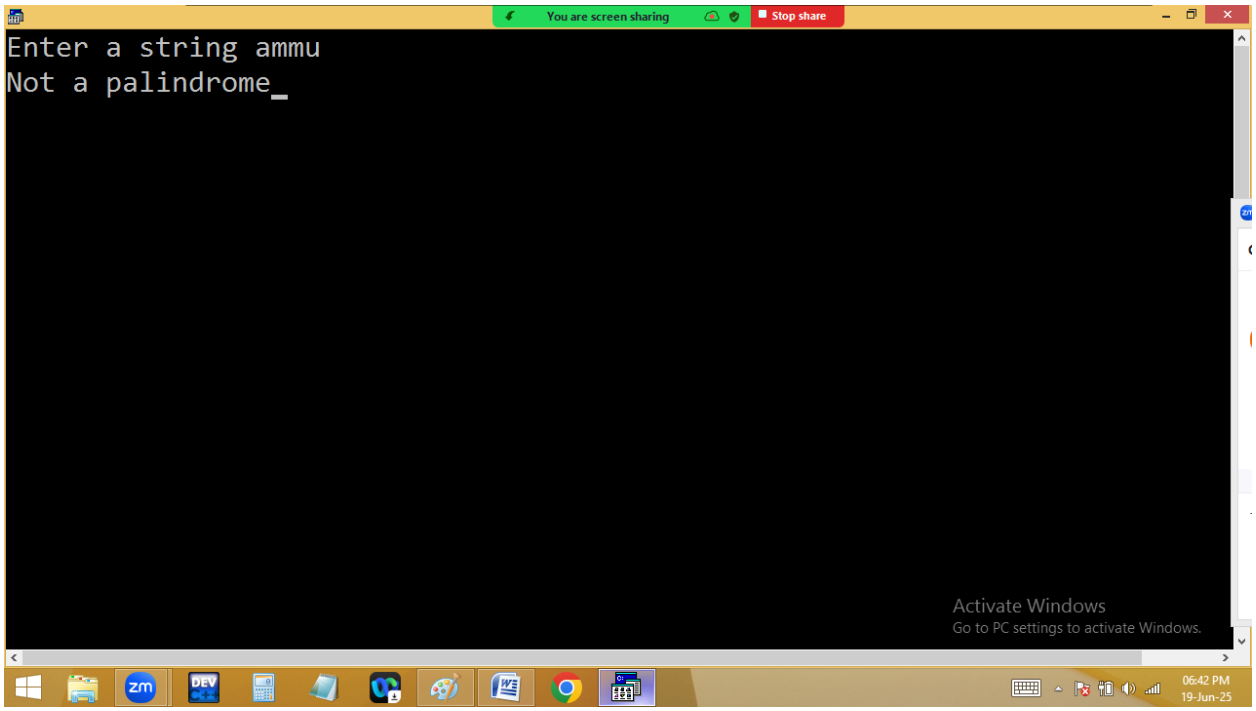
Eg: liril, madam, Malayalam, nitin, level, nayan,…


You are screen sharing
Stop share
Line 17 Col 53 Insert Indent Tab Fill Unindent * E:6PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
char s1[100],s2[100]; int i,j;
clrscr();
printf("Enter a string "); gets(s1);
for(i=0;s1[i]!='\0';i++)if(s1[i]>='A'&&s1[i]<='Z')s1[i]+=32;/*strlwr*/
for(i=0;s1[i]!='\0';i++); /* strlen */
for(i--,j=0;i>=0;i--,j++)s2[j]=s1[i]; /* rev + copy */
for(i=0;s1[i]!='\0';i++)
{
if(s1[i]!=s2[i]){printf("Not a palindrome");getch();return;}
}
puts("Palindrome");
getch();
}
Activate Windows
Go to PC settings to activate Windows.
06:41 PM
19-Jun-25


You are screen sharing
Stop share
Enter a string Akka
Palindrome
Activate Windows
Go to PC settings to activate Windows.
06:41 PM
19-Jun-25


You are screen sharing
Stop share
Enter a string 101
Palindrome
Activate Windows
Go to PC settings to activate Windows.
06:41 PM
19-Jun-25

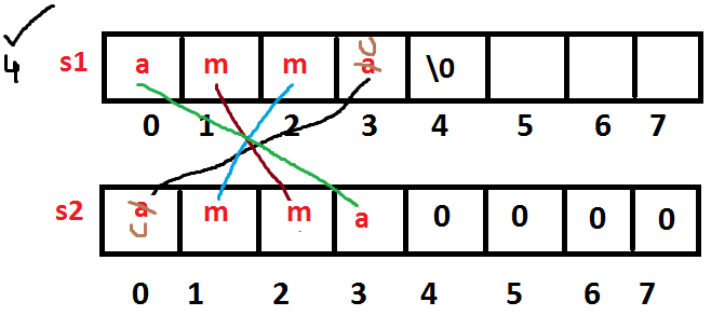

You are screen sharing
Stop share
Enter a string *$*
Palindrome
Activate Windows
Go to PC settings to activate Windows.
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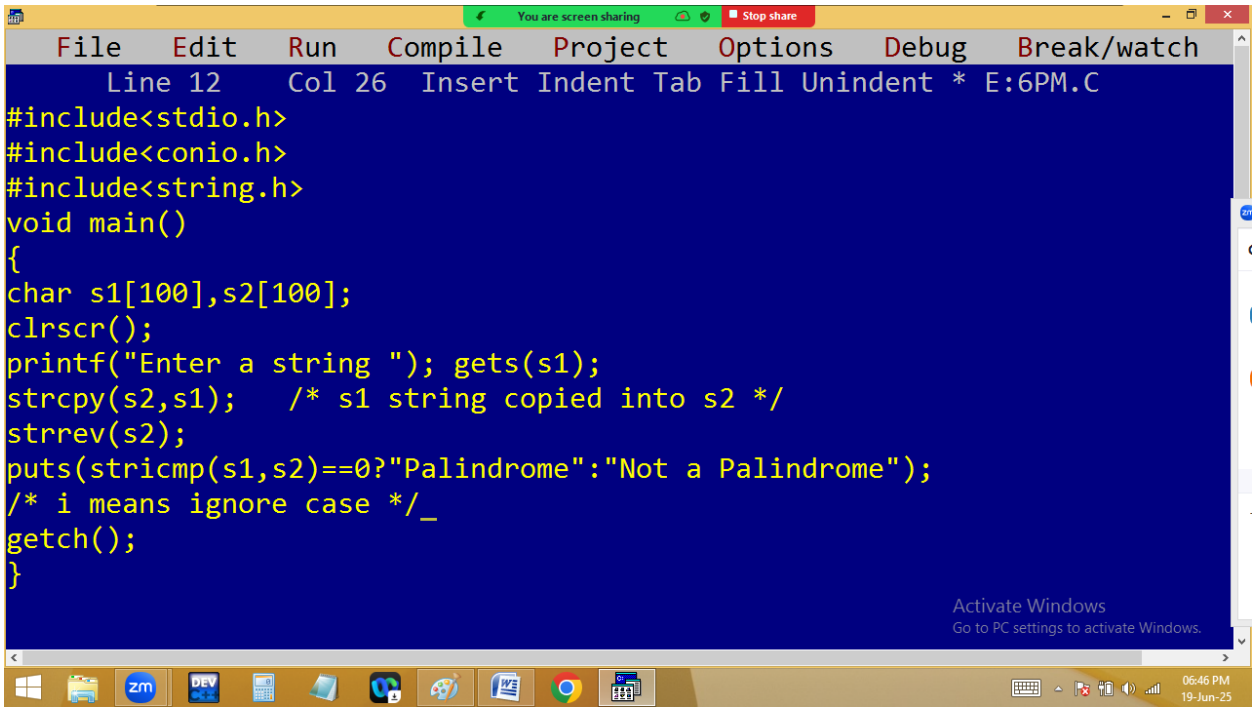

```
for( i=0; s[i]!='\0';i++); len

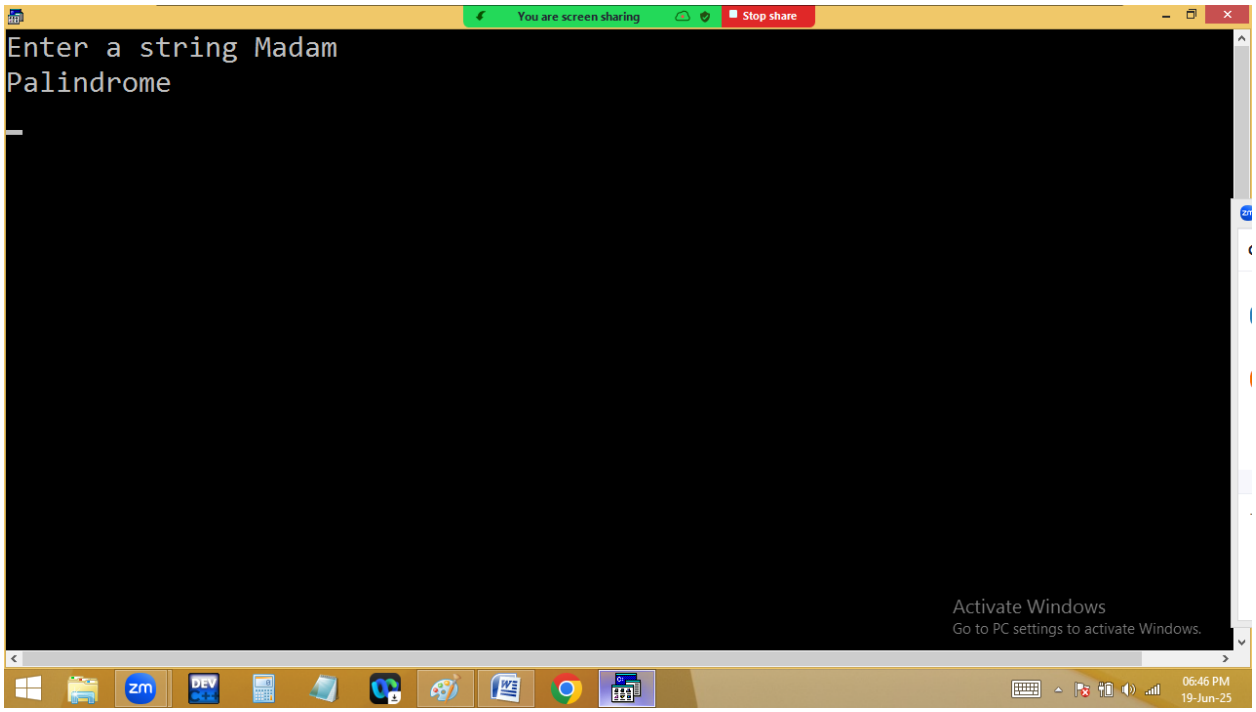
for(i--,j=0; i>=0;i--,j++)
    s2[ j]=s1[i]; rev+copy

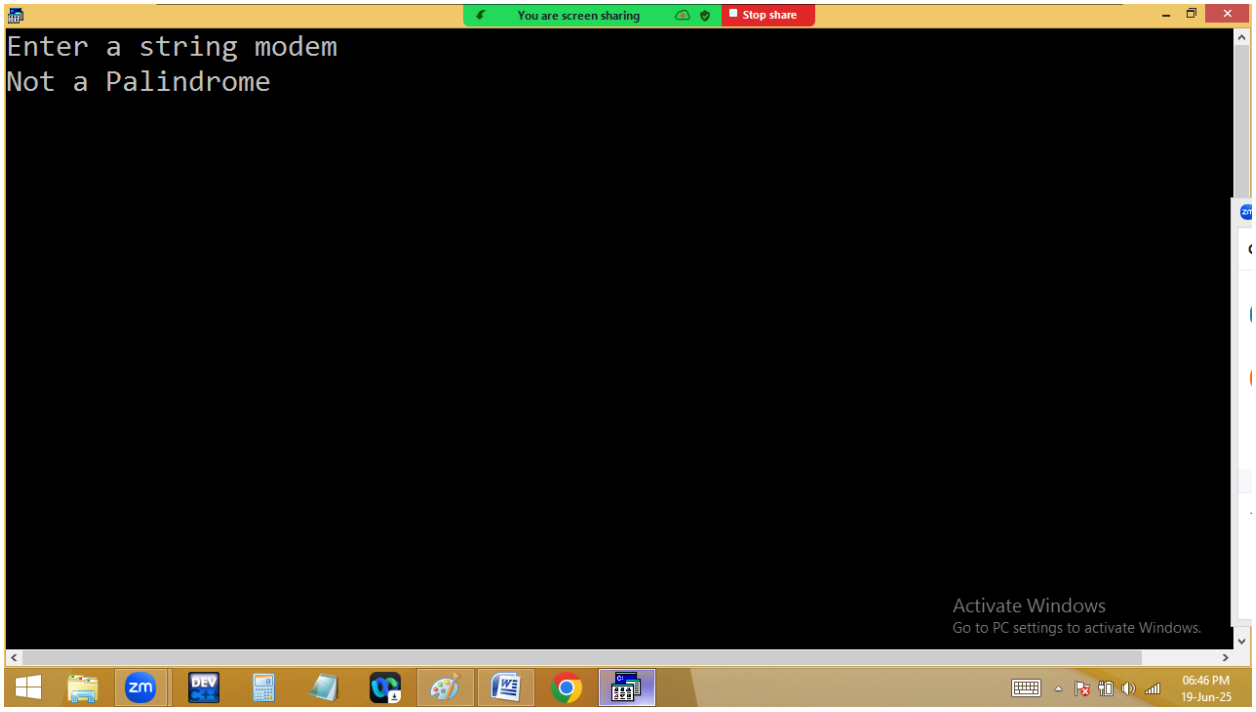
for(i=0;s1[i]!='\0';i++)
{
if(s1[i]!=s2[i]) strcom
    p("Not palin");return;
}
p( palindrome );
```



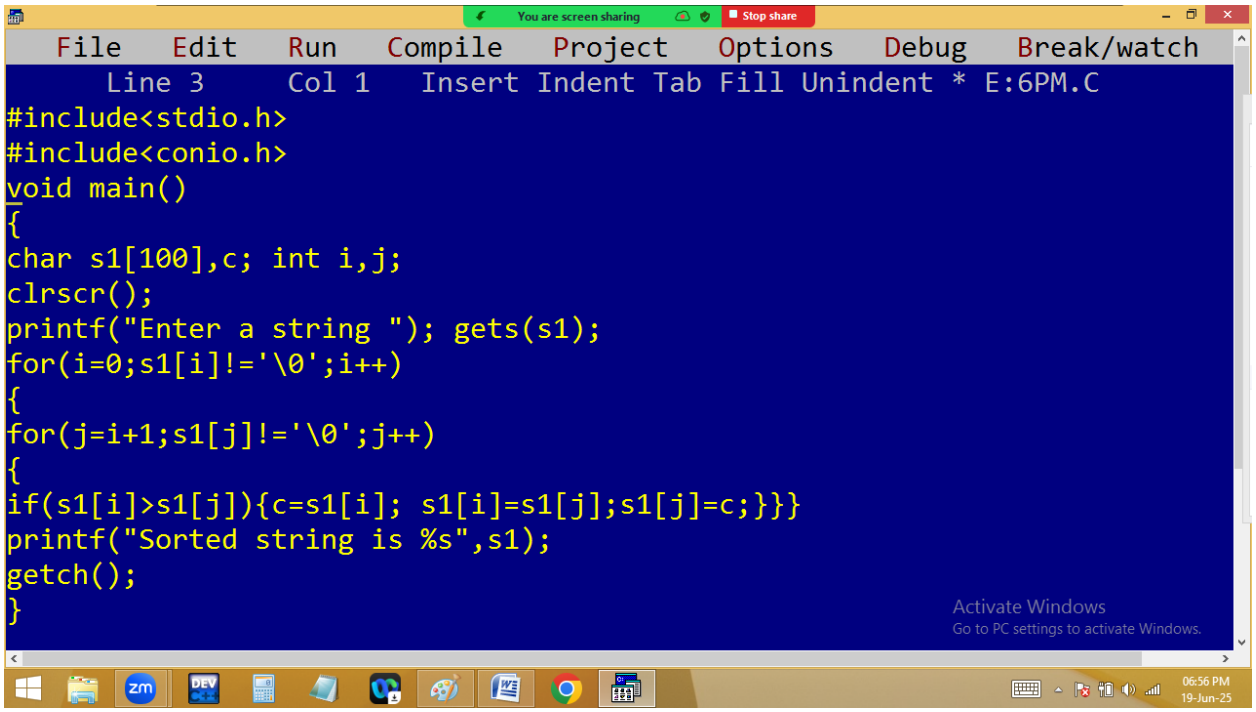
Using predefined functions:

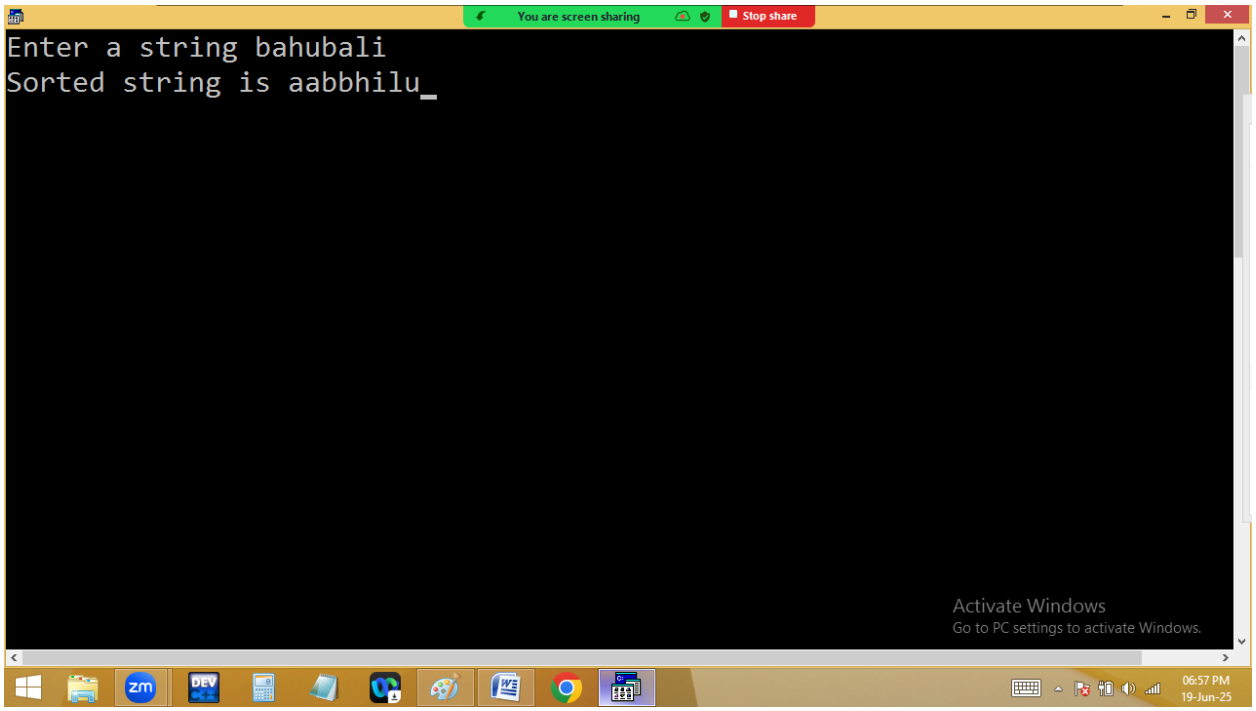

You are screen sharing
Stop share
File Edit Run Compile Project Options Debug Break/watch
Line 12 Col 26 Insert Indent Tab Fill Unindent * E:6PM.C
#include<stdio.h>
#include<conio.h>
#include<string.h>
void main()
{
char s1[100],s2[100];
clrscr();
printf("Enter a string "); gets(s1);
strcpy(s2,s1); /* s1 string copied into s2 */
strrev(s2);
puts(stricmp(s1,s2)==0?"Palindrome":"Not a Palindrome");
/* i means ignore case */_
getch();
}
Activate Windows
Go to PC settings to activate Windows.
06:46 PM
19-Jun-25

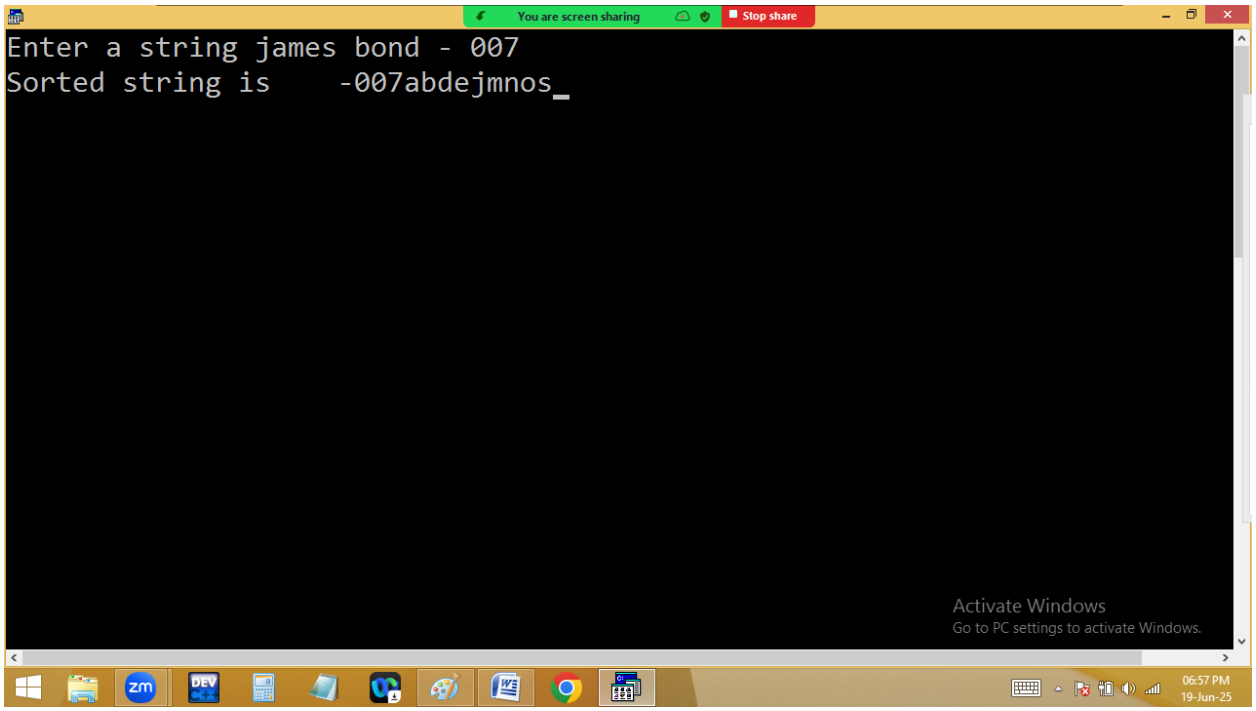

You are screen sharing
Stop share
Enter a string Madam
Palindrome
Activate Windows
Go to PC settings to activate Windows.
06:46 PM
19-Jun-25



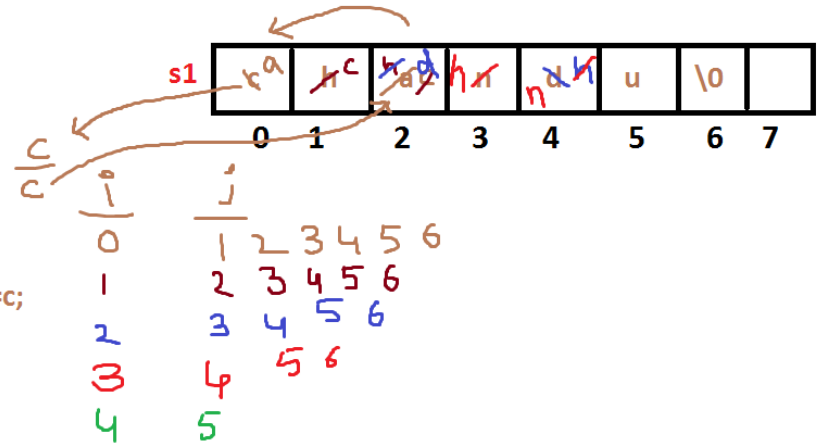
Sorting or string:



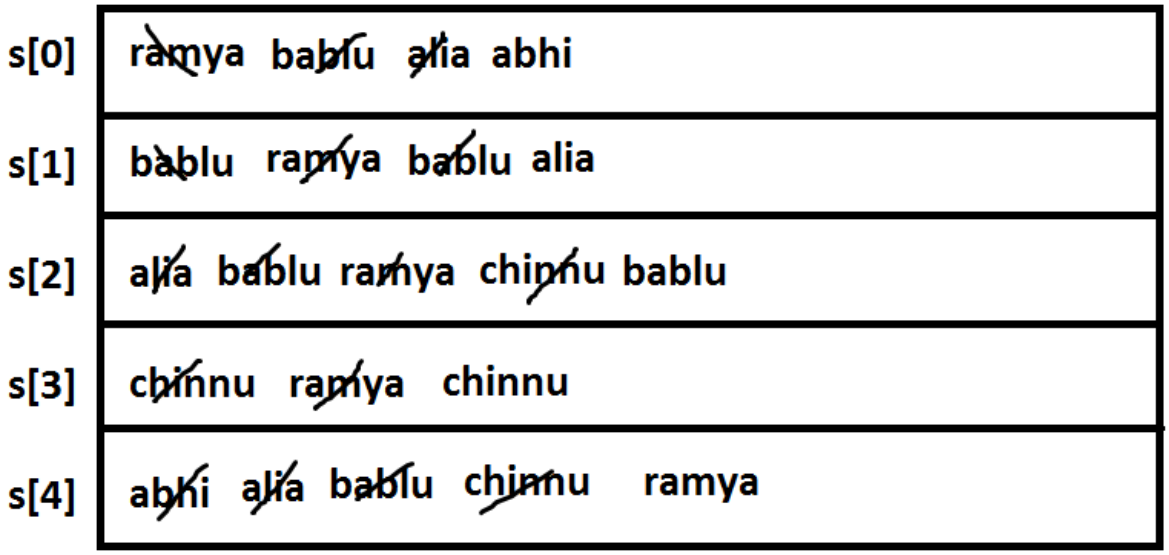

You are screen sharing
Stop share
Enter a string bahubali
Sorted string is aabbhilu
Activate Windows
Go to PC settings to activate Windows.
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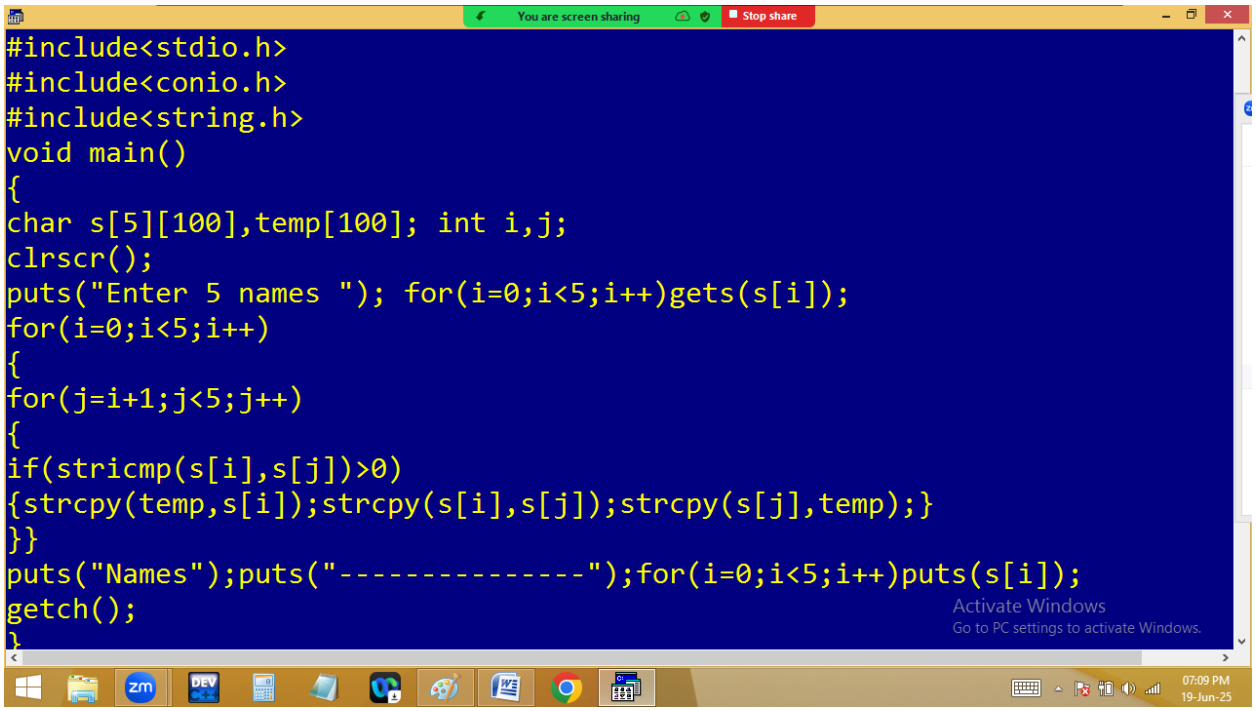


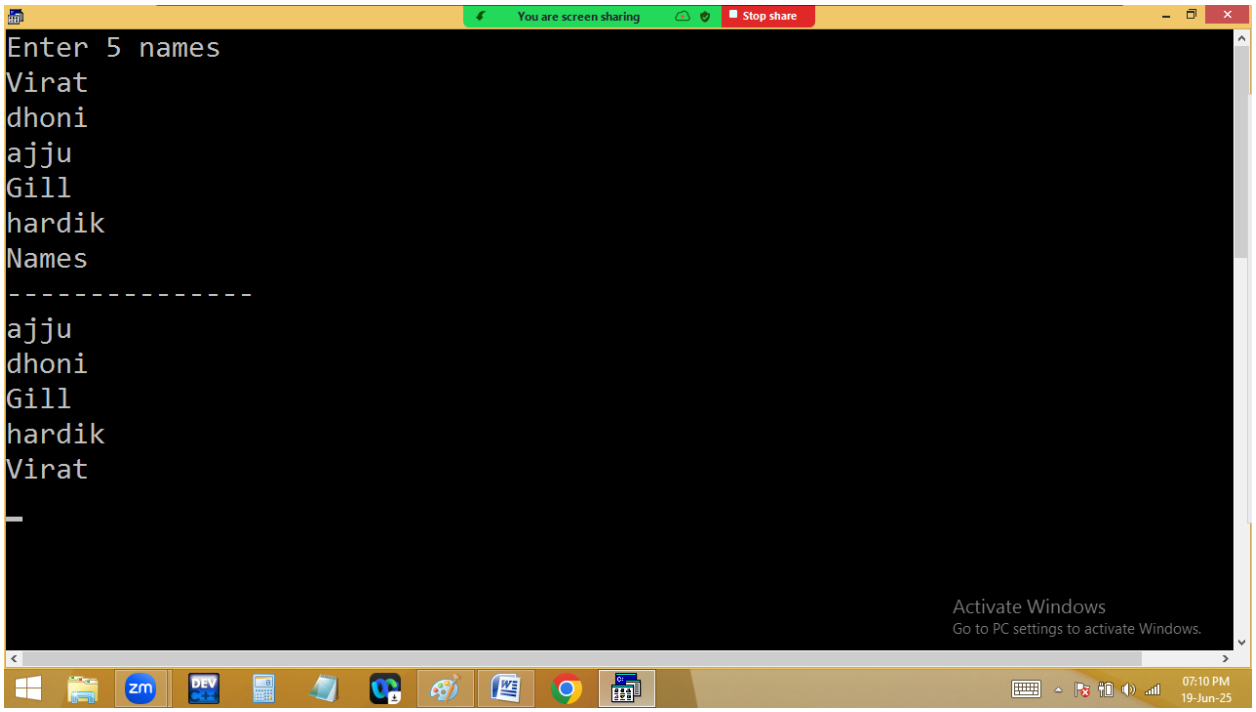

```
for(i=0; s1[i]!='\0';i++)
{
for(j=i+1;s1[j]!='\0';j++)
{
if(s1[i]>s1[j])
{
char c=s1[i]; s1[i]=s1[j];s1[j]=c;
}
```



Sorting of strings [selection sort]:


s[0]
ramya bablu alia abhi
s[1]
bablu ramya bablu alia
s[2]
alia bablu ramya chinnu bablu
s[3]
chinnu ramya chinnu
s[4]
abhi alia bablu chinnu ramya


You are screen sharing
Stop share
#include<stdio.h>
#include<conio.h>
#include<string.h>
void main()
{
char s[5][100],temp[100]; int i,j;
clrscr();
puts("Enter 5 names "); for(i=0;i<5;i++)gets(s[i]);
for(i=0;i<5;i++)
{
for(j=i+1;j<5;j++)
{
if(stricmp(s[i],s[j])>0)
{strcpy(temp,s[i]);strcpy(s[i],s[j]);strcpy(s[j],temp);}
}}
puts("Names");puts("---------------");for(i=0;i<5;i++)puts(s[i]);
getch();
}
Activate Windows
Go to PC settings to activate Windows.
07:09 PM
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You are screen sharing
Stop share
Enter 5 names
Virat
dhoni
ajju
Gill
hardik
Names
----
ajju
dhoni
Gill
hardik
Virat
Activate Windows
Go to PC settings to activate Windows.
07:10 PM
19-Jun-25

POINTERS

Pointer is a variable, which holds the address of another variable of same type.

Pointer is a memory location, which holds the address of another memory location.

Pointer is a derived data type.

Advantages:

1. Dynamic memory allocation.
2. Program performance is increased due to preventing memory wastage.
3. They are very much used in System programming.
4. They are very much used in dynamic linked list & Stacks [data structures].
5. **It allows to access local variable outside the function i.e. data sharing between functions.** [**call by address/Reference**].
6. **To handle strings, arrays etc in functions we need pointers.**

7. To handle **data files** we are using pointers.
8. They directly works on variable address. Due to this search time is reduced and execution speed is increased.

Dis-advantage:

They are not secured.